Count - distinct element

→ The Count distinct problem is the problem of finding the number of distinct elements in a data set stream with repeated elements.

Flajolet - Martin algorithm (FM)

(Find no. of unique elements)

→ This algorithm to find approximates the number of unique elements (objects) in a stream with a single pass.

→ Run time $O(n)$ time & needs $O(\log(m))$ memory.

Step:-

1. Find the hash function for given value x.
2. Enter the binary equivalent for every hash-function calculated.
3. Write the count of trailing zeros in each hash function bit.
4. Find the value of max count trailing zeros
5. Write the distinct value $R = 2^{\wedge}$

Example:

$S = \{1, 3, 2, 1, 2, 3, 4, 3, 1, 2, 3, 1\}$

$h(x) = (6x + 1) \bmod 5$

Assume $|b| = 5$ → that means 5 bits

x	h(x)	Reminder	Binary	Count Trailing Zeros	r = find max - no of Trailing Zeros	Distinct value $R = 2^r$
1	7	2	00010	1		
3	19	4	00100	2		
2	13	3	00011	0		
1	7	2	00010	1		
2	13	3	00011	0		
3	19	4	00100	2		
4	25	0	00000	-	2	$R = 2^2 = 4$
3	19	4	00100	2		
1	7	2	00010	1		
2	13	3	00011	0		
3	19	4	00100	2		
1	7	2	00010	1		

→ Trailing Zero's $\{1, 2, 0, 1, 0, 2, 2, 1, 0, 2, 1\}$

→ r = max [Trailing Zeros] = 2

→ output $R = 2^r = 2^2 = 4$

Counting 1's in a Stream

→ Given a Stream of 0's & 1's

→ It used to find the no of 1's in a Stream of data. (last k bits $k <= N$)

→ Commonly used Datar - Gianis - Indyk - motwani (DGIM) Algorithm.

→ Generaly it's called Motwani Algorithm.

→ use $O(\log^2 N)$ bits to represent a Window of N bits.

→ error rate is no more than 50%. (Solution gives approximate answer, not more than 50%

Elements:

1. Time Stamps:-

→ Each element entering in the Stream will be alloted a Time Stamp based on position of it

→ Example, if first bit has TimeStamp 1, then second bit will have TimeStamp 2, third bit 3 and so on.

2. Buckets :-

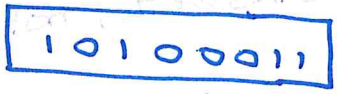
→ It used to represent time intervals in a data stream.

→ Algorithm divides the stream into buckets. each will have size of power of 2

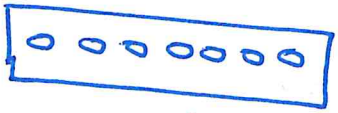
→ Buckets contains the bit 0 & 1.

Rules :-

1. Every bucket should contain atleast a Singee 1 in it.

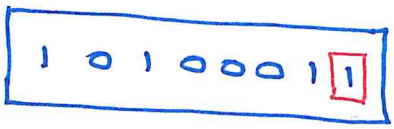


Valid bucket



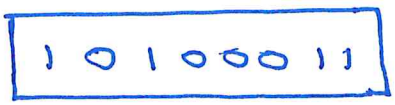
not Valid bucket

2. Right side of the bucket should Strictly start from 1





3. Length of the bucket is equal to the no. of 1's in it



Length of bucket = 4

Example:-

Consider the following stream

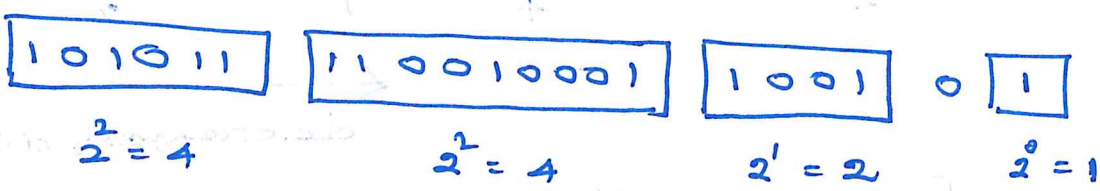
Stream = ... 1 0 1 0 1 1 1 1 0 0 1 0 0 0 1 1 0 0 1 0 1 ....

→ Stream goes like this from right to left

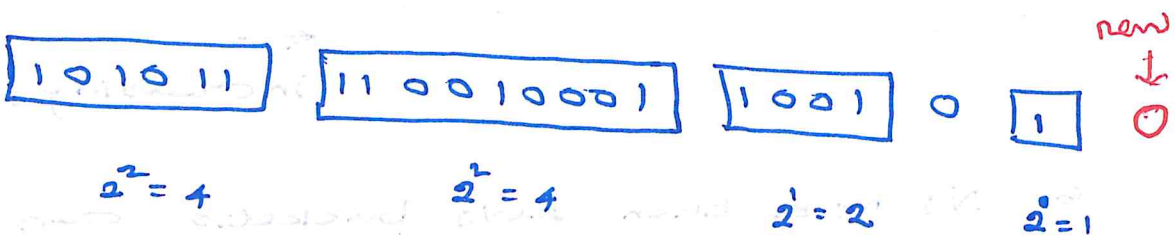
→ Add element in right hand side of stream

→ N = 20 (Consider 20 bits)

From bucket,

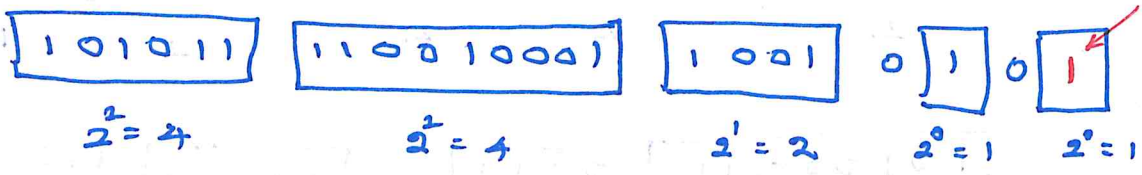


→ if new entering bit is '0', no change



No change the bucket.

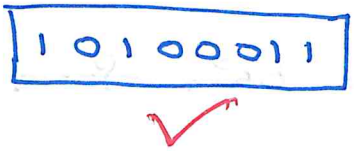
→ if new entering bit is '1'.



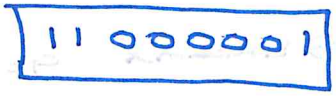
: After the bucket (one bucket added)

4. Every bucket length should be in power of 2.

ie. $2^0 = 1$ $2^1 = 2$ $2^2 = 4$ $2^3 = 8$ ....

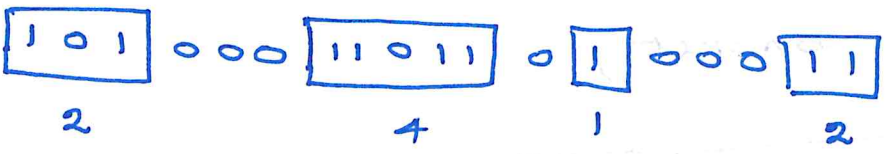


✓



✗

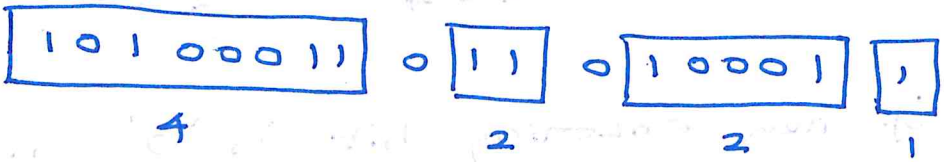
5. As we move to left, the bucket size should not decrease.



✗



decreasing order

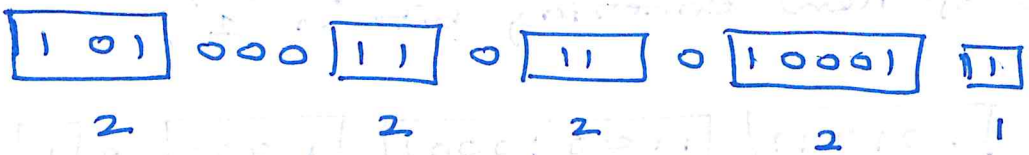


✓



increasing order

6. No more than two buckets can have same size.

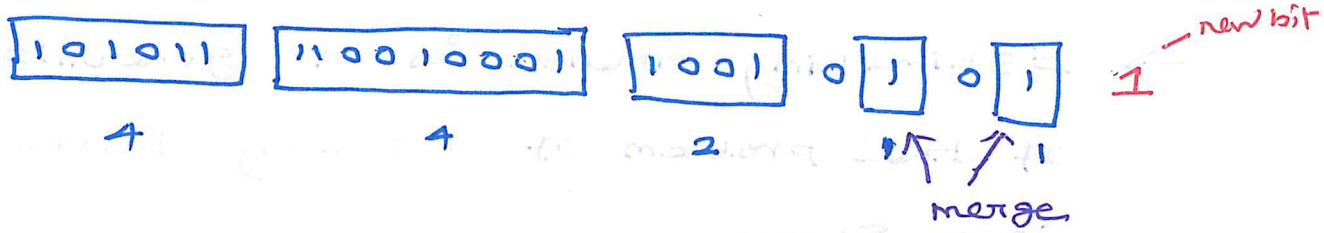


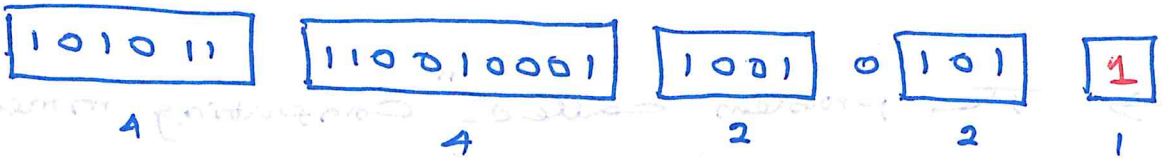
✗



✓

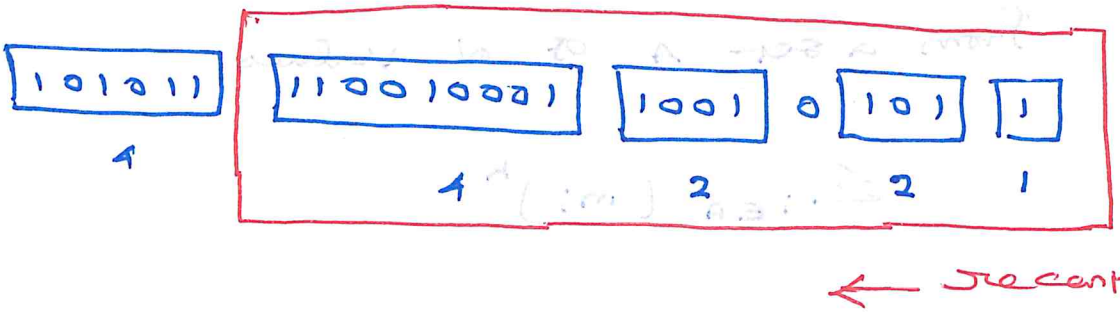
→ if new element entering is '1' after the bucket





Solved result :-

→ Count the no. of 1's in recent 18 bits



→ 4 + 2 + 2 + 1 = 9.

→ There are 9 bits of 1's in recent 18 bits.

—x—